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BE (5th Semester)

Examination, April-May, 2013

Branch : Mechanical, Mechatronics

MACHINE DESIGN - I

Time Allowed : Four Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) Read the instructions carefully before attempting.

(ii) Mention the reference taken from the data handbook as applicable.

Q. 1. (a) Define : **2**

(i) Stress concentration

(ii) Notch sensitivity

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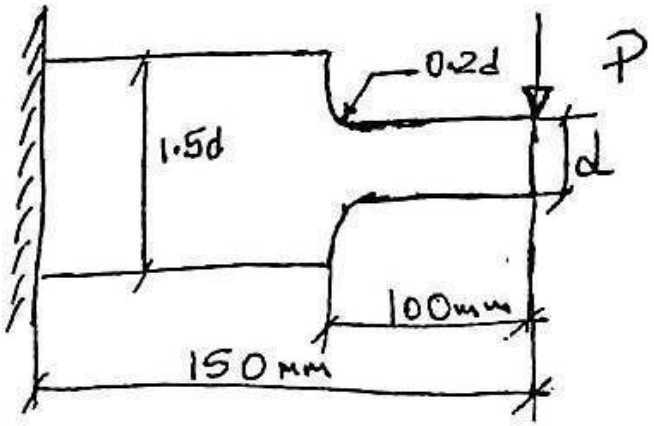
(b) Answer any one :

(i) A cantilever beam made of cold drawn steel (S_{ut} = 600 N/mm^2, S_{yt} = 380 N/mm^2). The force P acting on free end varies from –50N to +150N. Expected reliability is 90% and factor of safety is 2. Determine the diameter (q = 0.9). **14**



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(ii) Design a roller chain drive to connect a 15 kW, 1500 rpm electric motor to a centrifugal pump running at 750 rpm. The conditions of operation are constant load, fixed center distance, continuous lubrication, continuous running. 14

Q. 2. (a) List the various functions of keys. 2

(b) Answer any one :

(i) Design a joint to connect two rods of equal diameter to resist a tensile load of 60 kN. All parts of the joint are made of same material. 14

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Allowable stresses $\sigma_t = 60$ MPa, $\tau = 70$ MPa, $\sigma_c = 125$ MPa

(ii) Design and draw a cotter joint to support a load of 30 kN in compression and 30 kN in tension. The material used is carbon steel for which following allowable stresses may be used. 14

$\sigma_t = 50$ MPa, $\tau = 35$ MPa, $\sigma_c = 90$ MPa

Q. 3. (a) What is the difference between a rivet and a bolt. 2

(b) Answer any one :

(i) A wall bracket is attached to the wall by means of four bolts. The bracket is slipped about C. Allowable tensile stress

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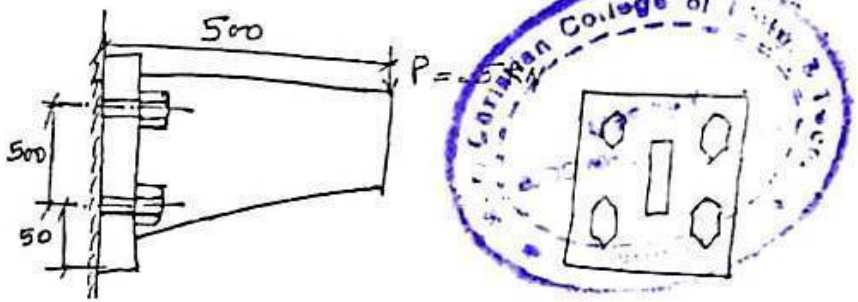
on the bolt is 35 N/mm^2 Determine the size of bolt using maximum principal stress theory. 14



(ii) A power screw having double start square thread of 25 mm nominal diameter & 5 mm pitch is added upon by an axial load of 10 kN. The outer and inner diameter of screw are 50 mm & 20 mm. The coefficient of thread friction & collar friction may be 0.2 and 0.15. The screw rotates at 12 rpm, allowable thread bearing pressure is 5.8 MPa. 14

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Find :

(1) Torque required

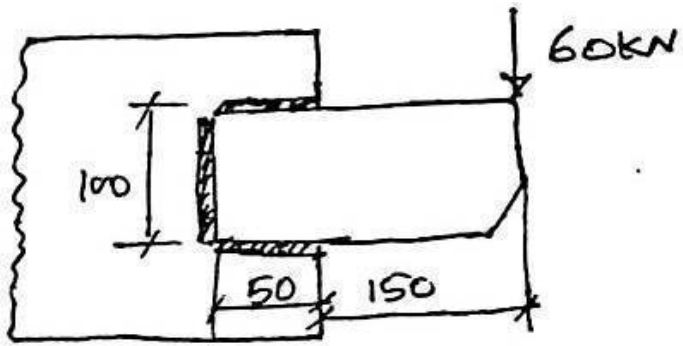
(2) Stress in screw

(3) Design the screw for ASME code

Q. 4. (a) What is a throat and leg in case of weld ? What is the relation between them ? 2

(b) Answer any one :

(i) A weld connection is subjected to eccentric force of 60 kN in the plane of weld. Determine the size of weld if the permissible shear stress is 100 MPa. 14



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(ii) Design a double riveted butt joint with two cover plates for a longitudual seam of boiler steel 1.5 m in diameter shell, subjected to steam pressure of 0.95 MPa. Assume joint efficiency to be 75%. 14

σ_t = 90 MPa, σ_c = 150 MPa, τ = 90 MPa

Q. 5. (a) What is strength, rigidity & stiffness in design of shaft. 2

(b) Answer any one

(i) A transmission shaft is supporting spur gear (B) and pulley (D). The shaft is mounted on bearings (A) and (C). The diameter of pulley and gear are 450 & 350 mm. The pulley transmits 20 kW power at 500 rpm to the gear. Assume $P_1 = 3P_2$ and $P_r = P_t \tan 20°$.

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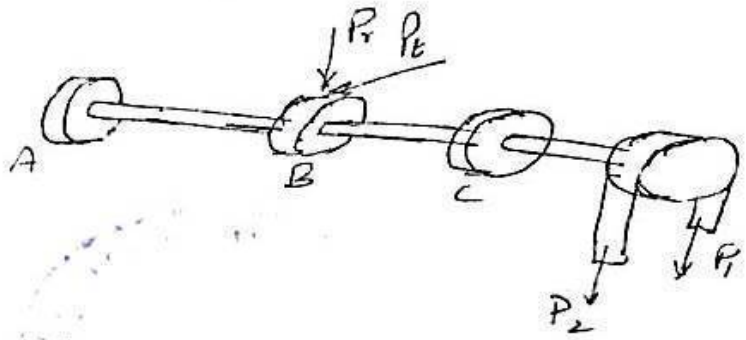
(S_{ut} = 700 N/mm^2, S_{yt} = 460 N/mm^2). The value of K_b and K_t are 1.5. Design the shaft using ASME code. 14



(ii) Design a muff coupling which is used to connect steel shaft transmitting 25 kW power at 360 rpm. The shaft and key are made of plane carbon steel 30 C8 (S_{yt} = S_{yc} = 400 MPa). The sleeve is made of grey cast iron FG 200 (S_{ut} = 200 N/mm^2). Factor of safety for key and shaft is 4 and for sleeve is 6. 14

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